

FLINTLOCK MUSKETS AND BREECH-LOADING RIFLES

Muzzle-loading flintlock muskets, such as the French Charleville and the British “Brown Bess”, proved reliable and robust. Thousands of these were made and remained in service until around 1840, by which time armed forces had recognized the ballistic advantages of rifled guns. The development of the unitary cartridge – which combined the bullet and ignition components in a single unit, and could be loaded by way of the breech – inspired new designs. Initially, many were conversions of muzzle-loading weapons, but these were followed by a range of purpose-built breech-loaders. The bolt action, pioneered by Johann von Dreyse and Antoine Alphonse Chassepot, was perfected by the Mauser brothers.

► **DREYSE NEEDLE GUN MODEL 1841**

Date	1841
Origin	Prussia
Weight	4.5kg (10lb)
Barrel	70cm (28in)
Calibre	13.6mm

Dreyse’s rifle had a simple straight-handed bolt, terminating in a needle that penetrated the length of a (linen) cartridge to detonate a percussion cap in the base of a Minié bullet.

▲ **BAKER RIFLE**

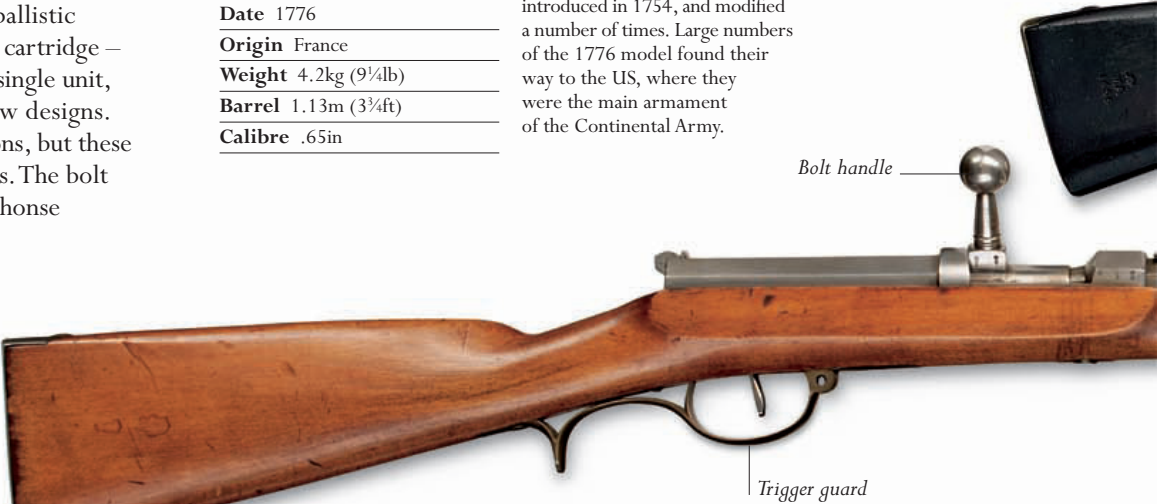
Date	1800–37
Origin	UK
Weight	4kg (8¾lb)
Barrel	76cm (30in)
Calibre	.625in

The rifle designed by English gunsmith Ezekiel Baker was chosen for riflemen in the British Army (see pp.192–93). Accurate to around 140m (460ft), it was an improvement over smoothbore muskets.

▲ **CHARLEVILLE MUSKET**

Date	1776
Origin	France
Weight	4.2kg (9¼lb)
Barrel	1.13m (3¾ft)
Calibre	.65in

Charleville muskets were introduced in 1754, and modified a number of times. Large numbers of the 1776 model found their way to the US, where they were the main armament of the Continental Army.



▲ **SHARPS PERCUSSION-CAP CARBINE**

Date	1848
Origin	US
Weight	3.5kg (7¾lb)
Barrel	45.5cm (18in)
Calibre	.52in

This percussion-cap breech-loader used a sliding breechblock to load a combustible cartridge, which was ignited by a tape primer or, in other models, a percussion cap.

